Residual Surrogates for Nonlinear DSGE Estimation: An Auditable Target-Corrected Workflow

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Abstract

Repeated nonlinear model solution can dominate likelihood-based estimation of dynamic stochastic general equilibrium models. We develop an auditable workflow that learns the discrepancy between a first-order reduced-order model and nonlinear extended-path solutions, evaluates the resulting profile inversion objective in JAX, and uses a cheap differentiable density to construct Hamiltonian proposals. A Metropolis test at each trajectory endpoint can preserve any declared target density that is evaluated there. This correction is established MCMC theory; our contribution is its integration with residual learning, parameter-dependent DSGE solution, cross-implementation checks, and a frozen evidence protocol. A preserved engineering pilot exercises the complete 23-equation nonlinear Galí model while estimating two policy parameters. It accepts 80 order-zero extended-path targets with maximum equation residual 9.92×10^{-9}, reduces held-out state error relative to the first-order model by 53.9 percent on average, and matches Python and JAX evaluations of the same surrogate objective within 1.71×10^{-13}. These results establish pipeline execution, not posterior fidelity or speedup: the pilot uses deterministic extended path, and its corrected HMC target remains a surrogate objective. We therefore state the confirmatory tests required before the method can support substantive inference claims.

1 Introduction

Likelihood-based estimation of nonlinear DSGE models repeats a difficult numerical task at many parameter values. Each likelihood evaluation can require a steady state, a local solution, nonlinear conditional trajectories, shock inversion, and measurement-error evaluation. The same work then appears inside optimization or posterior simulation. A neural surrogate is attractive only if it reduces this repeated cost without hiding the numerical failures that matter for inference.

This paper develops a workflow around that requirement. The method first solves the complete nonlinear equation system at a designed set of parameter, state, and shock inputs. It then fits a residual network to the gap between those solutions and a first-order reduced-order model (ROM). The ROM supplies global economic structure and a strong baseline; the network learns only the remaining nonlinearity. During posterior simulation, a cheap differentiable density can generate Hamiltonian trajectories, while a separately declared target density is evaluated at the endpoints. When the proposal map is reversible and volume preserving, the endpoint Metropolis test preserves that declared target.

The central contribution is the integrated estimation and validation design, not the generic observation that an approximate Hamiltonian may be corrected by an exact Metropolis test. That result is standard [Neal, 2011], and surrogate-gradient HMC has been studied with both random-basis and neural surrogates [Zhang et al., 2017, Li et al., 2019]. The contribution must instead be judged by four empirical questions. Does the residual surrogate improve on the ROM outside its training

parameter values? Does it preserve likelihood and posterior geometry in the inference-relevant region? Does endpoint correction target the density claimed in the paper? And does the complete workflow improve effective samples per second after training and compilation costs are counted?

We answer only the first, engineering-level version of these questions in the preserved pilot. The full confirmatory exercise is deliberately separated from the pilot. This separation prevents a successful software run from being mistaken for publication evidence.

2 Model, solution objects, and likelihoods

Let $\theta \in \Theta$ denote structural parameters, s_t predetermined states, x_t endogenous variables, and ε_t structural shocks. Write the nonlinear equilibrium system as

$$F(x_{t-1}, x_t, \mathbb{E}_t x_{t+1}, \varepsilon_t; \theta) = 0. \tag{1}$$

For a given parameter vector and shock design, a nonlinear extended-path solver produces a trajectory $x^F_{1:T}(\theta)$ from the full-order model (FOM). A first-order solution produces the cheaper trajectory $x^R_{1:T}(\theta)$. We fit a network r_ψ to the residual

$$\Delta x_t(\theta) = x^F_t(\theta) - x^R_t(\theta), \qquad \widehat{x}^F_t(\theta) = x^R_t(\theta) + r_\psi(z_t, \theta), \tag{2}$$

where z_t contains the states, shocks, and other features available at the prediction point. Parameter vectors, rather than individual rows, are assigned to training or validation. This prevents trajectories from the same parameter value from appearing on both sides of the split.

The target nonlinear posterior for data $y_{1:T}$ is

$$\pi_F(\theta \mid y) \propto p(\theta) L_F(y_{1:T} \mid \theta), \tag{3}$$

where L_F denotes an independently validated nonlinear likelihood with a declared treatment of latent shocks and every required change-of-variables term. The present implementation instead defines a surrogate inversion objective $\widetilde{L}_\psi$ by replacing nonlinear trajectories with (2), optimizing latent shocks, and evaluating the shock and measurement penalties at that optimizer. Unless an integration, Laplace adjustment, or required change-of-variables term is supplied, this is a profile or pseudo-likelihood objective rather than a marginal likelihood. The two objects are therefore scientifically distinct. Agreement between Python and JAX evaluations of $\widetilde{L}_\psi$ is implementation parity; it is not evidence that $\widetilde{L}_\psi = L_F$.

More explicitly, let $h_\psi(s_t, \varepsilon_t, \theta)$ map a state and structural shock into the observables implied by the ROM plus residual surrogate. At each date, the implementation computes

$$\widehat{\varepsilon}_t(\theta) = \arg\min_{\varepsilon} \left\{ \|R^{-1/2}[y_t - h_\psi(s_t, \varepsilon, \theta)]\|_2^2 + \|Q^{-1/2}\varepsilon\|_2^2 \right\}, \tag{4}$$

using a damped Gauss–Newton iteration, and then sums the Gaussian penalties evaluated at $\widehat{\varepsilon}_t(\theta)$. Here R and Q are diagonal in the pilot. The state passed to date $t+1$ follows the configured ROM or surrogate replay. Equation (4) makes the missing operation clear: the latent shocks are optimized, not integrated out. It also clarifies why agreement between two implementations of this calculation is a software check rather than likelihood validation.

The benchmark model is the complete nonlinear New Keynesian system distributed with the MacroModelling.jl examples and associated with the model in Galí [2015]. It contains 23 equations, 16 declared parameters, and three structural shocks. The reduced pilot estimates the Taylor-rule responses to inflation and output, ϕ_π and ϕ_y, while retaining every model equation. Reducing the parameter dimension makes a numerical posterior reference feasible; it does not turn the model into the familiar three-equation linearization.

2.1 Offline and online computations

The offline stage draws complete parameter vectors, solves a parameter-specific steady state and first-order ROM, and generates nonlinear extended-path targets conditional on designed state and shock paths. Each attempted target retains a case identifier, solver status, residual norm, and parameter-group identifier. Only the training partition determines feature scaling and network weights. Validation partitions are disjoint in complete parameter vectors, not merely in individual rows. The residual network and every preprocessing constant are then frozen and serialized with the model identity and source hashes.

The online stage transforms an unconstrained vector q to bounded structural parameters, adds the transform Jacobian to the declared log target, rebuilds the parameter-dependent objects required by the selected runtime, and evaluates either the profile objective or a nonlinear reference. A surrogate-gradient trajectory costs many cheap force evaluations but only one new target evaluation per transition. This accounting—rather than draws per second alone—is central when the endpoint target contains a nonlinear solution.

Extended path traces back to Fair and Taylor [1983]. Deterministic extended path sets future innovations to their conditional means and therefore does not capture Jensen effects in nonlinear expectations. Stochastic extended path approximates those conditional expectations over future shocks [Adjemian and Juillard, 2025]. This is why positive branching order, multiple nodes, and active variation in every shock are hard requirements for confirmatory evidence, even though order zero is useful for debugging deterministic nonlinearity.

3 Target-corrected surrogate trajectories

Let q be an unconstrained parameterization of θ, let p be Gaussian momentum with kinetic energy $K(p)$, and let $\pi_T(q)$ be the density the sampler is intended to preserve. A cheaper differentiable density $\pi_P(q)$ defines the force used in L leapfrog steps. Denote the resulting deterministic map by $(q', p') = \Phi_{\pi_P}(q, p)$. The acceptance probability is

$$\alpha\{(q,p),(q',p')\} = \min\left\{1, \exp\left[-H_T(q',p') + H_T(q,p)\right]\right\}, \quad H_T(q,p) = -\log \pi_T(q) + K(p). \tag{5}$$

Proposition 1 (Target invariance)**.** *Suppose Φ_{π_P} is reversible after momentum negation and volume preserving, and π_T is evaluated at both endpoints used in* (5)*. Then the Metropolis transition leaves π_T invariant, even when $\pi_P \neq \pi_T$.*

The proposition follows from the standard deterministic-proposal Metropolis argument for leapfrog HMC [Neal, 2011]. It places two practical restrictions on the implementation. First, the surrogate and every preprocessing transformation used in the force must be frozen during production sampling. Second, "exact" always modifies the declared target. If $\pi_T \propto p(\theta)\widetilde{L}_\psi$, the chain is exact for the surrogate-defined posterior, not for (3). Exact nonlinear posterior claims require endpoint evaluation of L_F itself.

For bounded parameters $\theta_j \in (a_j, b_j)$, the implementation samples $q_j \in \mathbb{R}$ and uses

$$\theta_j = a_j + (b_j - a_j)\,\mathrm{logit}^{-1}(q_j), \qquad \log J(q) = \sum_j \left[\log(b_j - a_j) + \log \sigma(q_j) + \log\{1 - \sigma(q_j)\}\right]. \tag{6}$$

Thus the endpoint target is defined in q coordinates and must include $\log J(q)$. Invalid steady states, indeterminate local solutions, failed nonlinear paths, or non-finite objective values must be rejected as points outside support. A large finite penalty is not equivalent: it creates an artificial plateau

with positive density. The preserved pilot still uses such a sentinel and therefore cannot support an exact structural-posterior claim.

The sampler has three implementation-level regression checks. When proposal and target densities coincide, its draws and accept/reject decisions match the ordinary static-HMC implementation under the same key. Under a deliberately shifted and rescaled proposal force, a fixed-seed multi-chain test recovers the known moments of a standard-normal target after endpoint correction. Finally, non-finite candidate targets receive zero acceptance probability. These tests check kernel plumbing; they do not replace convergence and posterior-fidelity evidence for the DSGE application.

This distinction is especially relevant for differentiable state-space and DSGE estimation, where automatic differentiation improves computation but does not resolve approximation error in the target [Childers et al., 2022]. Related work has made nonlinear DSGE likelihoods more tractable [Kollmann, 2017]; our evaluation therefore treats accuracy against an independently computed nonlinear reference as a primary outcome.

4 Validation design

The confirmatory design has two tiers. Tier 1 estimates (ϕ_π, ϕ_y) and uses a dense posterior grid. Grid resolution is doubled until posterior moments and interval endpoints stabilize. This tier isolates numerical correctness from high-dimensional MCMC behavior. Tier 2 estimates a prespecified four-to-six parameter subset and measures scaling with four independently initialized chains. The paper will report Tier 2 only after Tier 1 passes.

Training, validation, and posterior-region designs are disjoint in parameter vectors, shock sequences, and trajectories. Validation includes regular prior-interior points, posterior-region points, and a boundary-stress stratum. The last stratum is reported separately rather than deleted through a stability- margin screen. For stochastic extended path, evidence runs use positive branching order, multiple quadrature nodes, and nonzero coverage of every declared shock. Common case identifiers are evaluated by both the Python and MacroModelling.jl implementations.

The primary accuracy outcomes are scaled equilibrium residuals; held-out, per-output trajectory errors; paired improvement over the ROM with uncertainty; centered log-likelihood error; rank reversals near the mode; and normalized gradient error away from zero-gradient cases. Posterior comparison reports differences relative to pooled Monte Carlo error, interval endpoint differences, and a distributional diagnostic. Sampling reports rank-normalized split $\widehat{R}$, bulk and tail effective sample sizes (ESS), ESS per gradient evaluation, and ESS per second.

Timing uses the same workload, precision, and hardware for both methods. Compilation, host callbacks, device transfers, and training are reported separately. Warm likelihood microbenchmarks use repeated synchronized calls; end-to-end comparisons use independent processes. The current Schur QME path invokes a sequential SciPy host callback, so a GPU label does not imply a GPU-native likelihood. The available GPU-native matrix-sign alternative must pass Galí-specific solution and likelihood parity before it enters a speed claim.

The protocol is frozen before confirmatory execution. It records the repository revision, dirty-worktree state, environment, command, seeds, hardware, precision, and content hashes for every evidence artifact. Numerical failures remain in attempted-case denominators and are stratified by prior-interior, posterior-region, and boundary-stress designs. Machine-readable evidence is the source of manuscript tables; hand-entered numerical cells are not permitted.

Table 1: Preserved full-model Galí engineering pilots

Run	Target runtime	$N_\theta \times T$	Max. residual	Val. RMSE	Corrected	Accept.	Draws/s
Quick smoke	fixed-reference	4×4	7.21e-09	9.17e-05	No	1.000	0.378
Corrected resource	full-jax	4×4	7.21e-09	9.17e-05	Yes	1.000	0.0267
Default pilot	fixed-reference	8×10	9.92e-09	7.86e-06	No	1.000	8

Notes: All runs use the complete 23-equation nonlinear Galí source but estimate only two policy-rule parameters. N_θ is the number of parameter vectors and T is the observation horizon. Target construction used branching order zero and one node, so these are deterministic nonlinear extended-path pilots, not stochastic-expectations SEP validation. "Full-JAX" recomputes the parameter-dependent steady state and first-order ROM inside the surrogate profile inversion objective. It does not evaluate a full nonlinear SEP likelihood inside HMC. Correction therefore targets the declared surrogate objective, not an exact full-nonlinear posterior. The quick, corrected-resource, and default runs use respectively 1, 2, and 1 chains with 2, 2, and 50 post-warmup draws per chain. These runs establish execution and resource feasibility only.

5 Preserved engineering pilot

The pilot establishes four narrow facts. The parser, parameter-specific preflight, nonlinear equation solver, residual network, likelihood, and sampler compose on the full model. The order-zero extended-path targets satisfy a strict residual check in this small design. The fitted residual improves held-out state error over the first-order ROM. Finally, the Python and JAX implementations of the same surrogate profile inversion objective agree at the audited calibration.

The corrected resource run establishes only that endpoint evaluation is possible under a tiny budget. It retained four draws in total, achieved 100 percent acceptance, and processed 0.0267 draws per second. Those numbers are not convergence evidence and are not a speed advantage. The acceptance rate instead suggests that the static-HMC tuning is far from an efficient production regime.

6 Required confirmatory evidence

Table 2 defines the minimum evidence needed to promote the working draft to a submission. Counts are protocol defaults, not universal publication standards; learning curves, uncertainty intervals, and reference convergence determine whether they are sufficient.

All thresholds and case identifiers will be frozen before the final runs. The evidence builder reads raw result files, verifies the full-model identity, hashes its sources, and emits both machine-readable and manuscript inputs. A failed hard gate remains visible in the replication package; it is not converted into a missing observation or silently omitted path.

7 Limitations and interpretation

The preserved evidence is an engineering pilot. Its nonlinear targets use branching order zero and one node, so it does not validate stochastic expectations. Most likelihood calls hold the reference steady state and ROM fixed. The parameter-dependent full-JAX run still targets the surrogate profile inversion objective, assigns a finite -10^{12} sentinel to failed solves, and uses a host-callback Schur solve. An exact structural posterior instead requires explicit support handling and an economically defined likelihood. The sampler budgets are too small for convergence diagnostics, the data are synthetic and short, and no independent Julia trajectory or likelihood comparison is yet archived. The current evidence therefore supports neither exact nonlinear posterior sampling nor an end-to-end speedup claim.

Table 2: Hard gates for a credible reduced-model methods submission

Evidence family	Required result
Scope and provenance	Full 23-equation source verified by content hash and upstream revision; clean repository revision; complete environment, data, command, seed, and artifact manifests.
Nonlinear solution	Positive-order stochastic SEP with multiple nodes and all shocks active; raw attempted and successful counts; scaled residuals; regular, posterior, and boundary-stress strata.
Independent parity	Common case identifiers evaluated in Python and MacroModelling.jl; both failures retained in the denominator; QZ, trajectory, and likelihood errors reported on common successful cases.
Surrogate accuracy	Disjoint parameter, shock, and trajectory holdouts; per-output and tail errors; paired improvement over ROM with a 95 percent interval; learning-curve and out-of-distribution results.
Posterior fidelity	Two-parameter full nonlinear grid with resolution-doubling convergence; posterior moments, intervals, and distribution compared with the surrogate and corrected samplers using numerical or Monte Carlo uncertainty.
Sampling	Four tuned chains for the scaling experiment; rank-normalized split $\widehat{R}$, bulk and tail ESS, failures, ESS per gradient evaluation, and ESS per second.
Timing	Same-work, same-precision comparison; synchronized warm calls and independent end-to-end processes; compilation, transfers, training, host callbacks, and peak memory reported separately.

Notes: These are claim-specific design requirements. Prespecified thresholds and case lists must be frozen before confirmatory execution. Passing them supports the reduced Galí experiment only; it does not establish universal accuracy across DSGE models.

These limitations determine the paper's next experiments rather than merely qualifying its conclusion. A credible methods result requires the two-parameter reference experiment to pass before scaling the posterior dimension. If the surrogate does not preserve the grid posterior, or if its full accounting does not improve ESS per second, the negative result should be reported and the claim reduced to numerical infrastructure.

8 Conclusion

Residual learning can be useful for nonlinear DSGE estimation because a first-order model provides a disciplined baseline and the network concentrates on the nonlinear discrepancy. Target-corrected HMC can remove error from the trajectory force, but only relative to the density actually evaluated at the Metropolis endpoints. The preserved Galí pilot shows that the software path is feasible and identifies the dominant remaining costs. Publication readiness now depends on stochastic-SEP validation, an independently converged nonlinear posterior reference, cross-language parity, efficient multi-chain sampling, and same-work timing. The accompanying evidence protocol makes those requirements falsifiable and reproducible.

Disclosure on computational assistance

Automated coding and language tools were used to inspect software, generate tests, and edit the working draft. Every numerical claim in the paper is linked to a machine-readable source file and

must be verified by the authors. The final submission will follow the target journal's disclosure policy.

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